

1. Overview

Quillbase is a distributed key-value store written in Rust. This handbook describes release 4.2 and is written for operators who run production clusters. Quillbase is a fictional product invented to demonstrate retrieval evaluation.

A cluster is made of nodes that share membership information through a gossip protocol and keep data consistent with a replicated log. Any node can accept a client request and forward it to the right owner.

2. Network ports and TLS

Clients connect to a node on TCP port 7421 by default. The administration interface listens on port 7422 and must never be exposed to the public internet. Nodes exchange gossip messages with each other on port 7430.

Every connection that does not come from the loopback interface must use TLS 1.3. Plain-text connections from other hosts are refused, and the refusal is written to the audit log.

3. Storage engine

Quillbase writes every change to a write-ahead log before it is applied to the data files. The log is split into segments of 64 MiB each, and old segments are archived once every replica has applied them.

By default the log is flushed to disk every 5 ms using group commit, which trades a small delay for much higher throughput. Data files use a block size of 16 KiB. Background compaction starts when 4 overlapping tables exist at the same level.

4. Replication

Each key is stored with a replication factor of 3 by default, so three different nodes hold a copy of it. A write is acknowledged to the client once 2 of 3 replicas have confirmed it, which is called a quorum write.

Read repair is enabled by default: when a read notices that two replicas disagree, the stale copy is fixed in the background. Operators should raise an alert when cross-region replication lag stays above 800 ms.

5. Snapshots and restore

Snapshots run every night at 02:30 UTC and are kept for 14 days. Each snapshot has an identifier that starts with the prefix `qb-snap-` followed by the date and a short hash.

To restore a node, run `qbctl restore` with the snapshot identifier. Restoring replaces all local data on the node, so drain the node from the load balancer first and let replication refill it afterwards.

6. Memory

The block cache uses 25% of system memory by default. The setting `cache_ratio` can raise it, but the hard limit is 70% of memory and larger values are rejected at startup.

Leave the rest of the memory free for the operating system page cache. Running the cache close to the hard limit makes compaction slower and increases the risk of the process being stopped by the kernel.

7. Security and access

API tokens stay valid for 12 hours and must then be renewed by the client. Three roles exist: reader, writer and operator. Only the operator role may change cluster settings or trigger a restore.

The audit log records every administrative action together with the token that performed it. Audit entries are retained for 90 days and can be exported as JSON lines.

8. Upgrades

Clusters are upgraded with a rolling upgrade that touches one node at a time. Wait at least 5 minutes between nodes so that replication lag can return to normal before the next node restarts.

Downgrading across a major version is not supported. Always take a snapshot before an upgrade and check the release notes for changes to the on-disk format.

9. Monitoring

The metric `qb_commit_latency_p99` tracks the 99th percentile commit latency. The default alert fires when it stays above 25 ms for ten minutes.

The bundled dashboard is called Heronwatch. It shows replication lag, cache hit ratio and the compaction backlog for every node in one view.

10. Error codes

QB-4091 means that quorum was lost because too many replicas are unreachable. QB-5003 means that the data disk is full and writes are rejected until space is freed. QB-2210 means that the API token has expired and must be renewed.

11. Limits

The maximum key size is 1 KiB and the maximum value size is 8 MiB. A single batch may contain at most 500 keys. A cluster supports up to 256 namespaces.

12. Support

The support desk is open on weekdays from 08:00 to 18:00 UTC. A Severity 1 incident, which means a full outage, receives a first response within 30 minutes. Severity 2 incidents receive a response within 4 hours.

Appendix A. Configuration reference

`gossip_interval`: Time between gossip rounds. Default value: 200 ms.

`heartbeat_timeout`: Time before a peer is suspected. Default value: 1500 ms.

`max_connections`: Client connections per node. Default value: 4096.

`idle_timeout`: Idle client connection limit. Default value: 300 seconds.

`read_timeout`: Deadline for a read request. Default value: 2000 ms.

`write_timeout`: Deadline for a write request. Default value: 3000 ms.

`cache_shards`: Number of block cache shards. Default value: 16.

`bloom_bits_per_key`: Bloom filter size per key. Default value: 10 bits.

`query_log_retention`: How long query logs are kept. Default value: 7 days.

`metrics_retention`: How long metrics are kept. Default value: 30 days.

`wal_archive_retention`: How long archived log segments are kept. Default value: 3 days.

`max_replication_streams`: Parallel replication streams. Default value: 8.

`rebalance_rate`: Data moved per second during a rebalance. Default value: 48 MiB.

`token_renewal_window`: How early a token may be renewed. Default value: 15 minutes.

`login_attempt_limit`: Failed logins before lockout. Default value: 6.

`lockout_duration`: Length of an account lockout. Default value: 20 minutes.

`tls_cert_reload`: Interval for reloading certificates. Default value: 60 seconds.

`audit_flush`: Interval for flushing the audit log. Default value: 10 seconds.

`compaction_threads`: Background compaction workers. Default value: 2.

`snapshot_compression`: Snapshot compression level. Default value: zstd level 3.

`batch_timeout`: Deadline for a batch request. Default value: 4000 ms.

`gossip_fanout`: Peers contacted per gossip round. Default value: 3 peers.

`hint_ttl`: How long hinted handoffs are stored. Default value: 2 hours.

`scan_page_size`: Rows returned per scan page. Default value: 1000.

`ttl_sweep_interval`: How often expired keys are removed. Default value: 90 seconds.

`checkpoint_interval`: Time between checkpoints. Default value: 10 minutes.

`max_open_files`: File descriptors per process. Default value: 65536.

`slow_query_threshold`: Latency that marks a slow query. Default value: 750 ms.

`quota_check_interval`: How often namespace quotas are checked. Default value: 45 seconds.

`log_rotation_size`: Size of an application log file. Default value: 200 MiB.

`dns_refresh`: Interval for refreshing peer addresses. Default value: 120 seconds.

`election_timeout`: Time before a leader election. Default value: 3500 ms.

`range_split_size`: Size at which a range is split. Default value: 512 MiB.

`range_merge_size`: Size below which ranges are merged. Default value: 64 KiB.

`client_retry_limit`: Retries performed by the client library. Default value: 3.

`backoff_base`: Starting delay for retry backoff. Default value: 40 ms.

Appendix B. Error code catalog

QB-1001 means that the configuration file could not be parsed.

QB-1007 means that the unknown setting name in the configuration.

QB-1013 means that the certificate file is missing or unreadable.

QB-2101 means that the namespace does not exist.

QB-2102 means that the namespace name is not valid.

QB-2144 means that the request is not allowed for this role.

QB-2301 means that the key is larger than the key size limit.

QB-2302 means that the value is larger than the value size limit.

QB-3012 means that the node is draining and refuses new writes.

QB-3040 means that the range is being split, retry shortly.

QB-3077 means that the hinted handoff queue is full.

QB-4010 means that the replica did not answer within the write timeout.

QB-4022 means that the read repair was skipped under load.

QB-4108 means that the leader election is in progress.

QB-4150 means that the clock skew between nodes is too large.

QB-5011 means that the compaction backlog is above the safe level.

QB-5020 means that the block cache could not be allocated.

QB-5044 means that the log segment checksum mismatch.

QB-6001 means that the snapshot upload to object storage failed.

QB-6014 means that the snapshot identifier was not found.

QB-6020 means that the snapshot belongs to a different major version.

QB-7003 means that the audit log export is already running.

QB-7010 means that the operator token required for this command.

QB-8002 means that the client library version is too old.